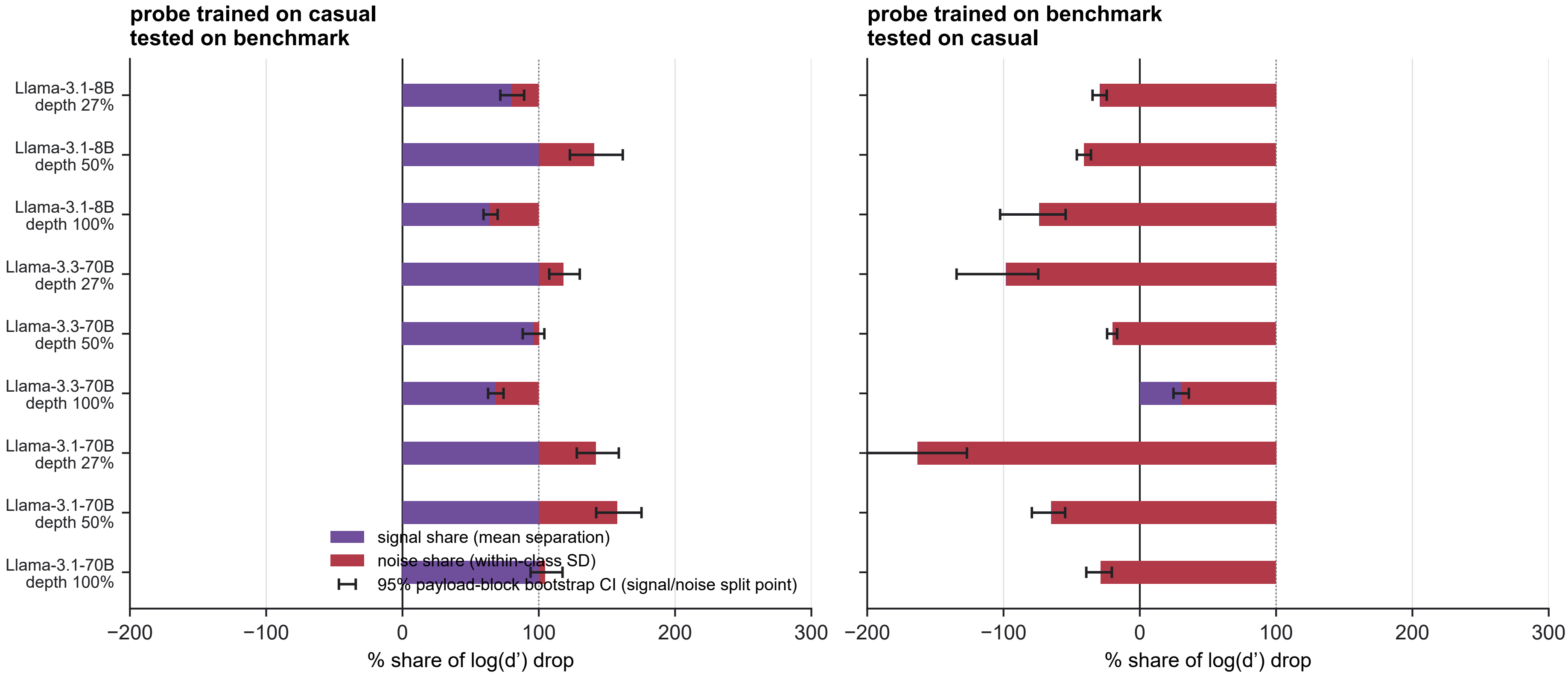
Cross-format transfer loses discriminability via two different failure modes, by direction



Exact decomposition of d' (not a correlation, and not claimed as an exact AUC decomposition): log(d'_self / d'_transfer) splits additively into a signal-loss term and a noise-growth term. Real held-out ROC AUC is reported per cell as EXTERNAL VALIDATION that d' tracks true performance, not as a quantity the algebra decomposes exactly. Casual→benchmark transfer loses discriminability almost entirely via a SIGNAL-LOSS failure mode (the probe direction stops tracking the true separation in the new format). Benchmark→casual transfer loses discriminability almost entirely via a NOISE-GROWTH failure mode (signal is usually preserved or even improves; within-class scatter along the same direction explodes). Replicated at 3 depths (fixed 27% anchor, mid, final layer) in all 3 models. Descriptive failure-mode characterization, not a claim about the underlying causal neural mechanism. Error bars: 95% CI on the signal/noise split point, 1,000-rep payload-block bootstrap; in all 9 model×layer combinations the CI stays entirely on the dominant side (never crosses 50%).